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UAV-mounted electrochemical soil probing system for geospatial phosphorus mapping

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ABSTRACT

Accurate spatial mapping of soil phosphorus remains challenging due to the high cost and limited spatial density of laboratory soil analyses and the poor performance of remote-sensing approaches for phosphorus estimation. This study presents a UAV-mounted soil sensing system that integrates autonomous mechanical probing with in-situ electrochemical measurement and spatial machine learning to enable high-resolution phosphorus mapping under field conditions. The platform combines a lightweight drone-deployable probing mechanism, an ion-selective electrochemical probe, onboard control electronics, and wireless telemetry to collect GPS-referenced soil nutrient measurements during automated flight operations. Field experiments were conducted at two contrasting agricultural sites: a vineyard in British Columbia, Canada, and an intensively managed flower farm in Arusha, Tanzania. Three complementary datasets were generated to evaluate spatial nutrient structure at both location and within-location scales. Spatial feature engineering incorporated positional geometry and neighborhood statistics, and spatial autocorrelation was assessed using Moran's I and variogram analysis. Random Forest classifiers were trained using spatially grouped cross-validation to prevent performance inflation due to spatial dependence.

Across study sites, the proposed approach achieved weighted F1-scores ranging from 0.82 to 0.91, representing improvements of 25–40% over baseline classifiers. The resulting phosphorus maps were spatially coherent and agronomically plausible, successfully identifying low-phosphorus zones in the Canadian vineyard despite limited nutrient variability, and distinct low- and high-phosphorus regions in the Tanzanian field where fertility gradients were stronger. Analysis of feature importance highlighted the dominant role of local spatial contrast and neighborhood information, underscoring the importance of explicitly modeling spatial context when using in situ electrochemical sensors. The results demonstrate that coupling UAV-based automated soil probing with spatially informed machine learning provides an effective pathway for transforming sparse physical measurements into continuous phosphorus distribution maps. This integrated sensing-to-mapping framework offers a scalable foundation for site-specific nutrient management and can be extended to additional soil properties and precision agriculture applications.

1. Introduction

Efficiently monitoring soil nutrients is fundamental to precision agriculture. Timely and accurate assessment of soil fertility influences crop yield, fertilizer use and long-term environmental health. Nitrogen (N), phosphorus (P), and potassium (K) are primary macronutrients essential for plant growth, yet they exhibit spatially heterogeneous distributions with moderate to strong spatial dependence across field scales, as quatified by variograms (Zhang et al., 2019). Conventional laboratory methods such as ICP-OES (inductively coupled plasma optical emission spectroscopy) and colorimetric assays provide high accuracy but are costly, time-consuming and spatially sparse, limiting their value for routine high-density monitoring (Viscarra Rossel and Bouma, 2016). These limitations often result in uniform fertilizer application, raising input costs and contributing to eutrophication in surface waters (Xia et al., 2020).

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Among the primary macronutrients, phosphorus presents a uniquely high-stakes management challenge. Unlike nitrogen, which can be replenished through atmospheric fixation and exhibits rapid cycling, phosphorus is a finite resource with no gaseous phase and extremely limited mobility in soils (Solangi et al., 2023). Phosphorus losses through runoff or erosion are effectively irreversible at field scales, while accumulated legacy phosphorus can persist for decades, continuing to impair water quality even after fertilizer inputs are reduced (Solangi et al., 2023). Globally, phosphorus mismanagement is a leading driver of freshwater eutrophication, harmful algal blooms, and regulatory pressure on agricultural producers (Daniel et al., 1998). At the same time, phosphorus deficiency directly constrains root development, early plant vigor, and yield potential, particularly in intensively managed systems (Hopkins and Hansen, 2019). This combination of agronomic necessity, environmental risk, and long-term persistence makes phosphorus the most critical and unforgiving nutrient to manage accurately at fine spatial scales.

UAVs have become increasingly common in agriculture, evolving from observational tools into platforms capable of data acquisition and decision support. Adoption is rising even in regions dominated by family-operated farms, with UAV ownership projected to reach 30% in small holder operations by 2027 (Kuehl et al., 2025). As UAV ownership grows, there is increasing interest in expanding their roles beyond imaging into soil and nutrient diagnostics.

Recent work in UAV multispectral and hyperspectral imaging has achieved strong performance for nitrogen estimation and soil organic matter mapping (Sun et al., 2022; Yang et al., 2021). CNN-based approaches now outperform traditional regressions for soil organic carbon, nitrogen and phosphorus (Zhang et al., 2025), and hyperspectral datasets can capture broad NPK variability (Misbah et al., 2022). However, these imaging methods remain indirect. They correlate well with nitrogen but perform poorly for phosphorus and potassium because canopy-level spectral signatures do not reflect soil availability reliably (Mazur et al., 2023).

Other recent work on fully autonomous unmanned aerial vehicles has shown that robust navigation models can enable precise horizontal mapping over extensive regions by combining waypoint-based and autonomous flight planning, significantly reducing operator-induced errors and enabling consistent, repeatable coverage of large areas. Building on this paradigm, the proposed UAV-mounted electrochemical soil probing system targets similarly autonomous, high-resolution geospatial mapping of soil phosphorus across agricultural fields (Bakirci, 2024).

Unlike nitrogen and chlorophyll-related indicators, phosphorus availability does not produce a direct spectral signature that can be reliably detected through UAV-based optical imaging. Its effects are usually expressed indirectly through plant stress responses, which can also arise from other nutrient limitations, water stress, disease, or environmental variation. As a result, remote sensing approaches often show weak and inconsistent relationships with actual soil phosphorus availability. In this work, the UAV is not used as a spectral phosphorus sensor, but rather as an automated deployment and geospatial sampling platform for direct electrochemical soil probing, enabling spatially distributed in situ phosphorus measurements.

Phosphorus is therefore particularly challenging. It is agronomically essential yet environmentally sensitive. Deficiency limits crop growth, while overapplication increases runoff risk and eutrophication (Khan et al., 2023). Unlike nitrogen, phosphorus has no gaseous phase and remains mostly immobile in soil, producing highly heterogeneous spatial patterns (Khan et al., 2023). These constraints highlight the need for direct measurements. Focusing on phosphorus in this study addresses a nutrient that imaging methods struggle to quantify but that strongly influences both productivity and environmental stewardship (Khan et al., 2023).

Parallel progress in ground-based sensing has introduced portable electrochemical devices for nitrate and phosphate detection (Ali et al., 2019; Baumbauer et al., 2022; Patel et al., 2022). Reviews confirm the maturity of modern electrochemical sensing for real-time macronutrient detection (Dattatreya et al., 2024; Hossain et al., 2024). Microfluidic technologies have also demonstrated field-ready nutrient analysis with high sensitivity (Lu et al., 2024; Mishra et al., 2025). Complementary IoT-enabled systems have been developed for decision support (Mohanty et al., 2024; Munezero et al., 2024). Despite these advances, most sensors require manual deployment, which limits scalability for continuous field mapping. Commercial ion-selective electrode probes provide rapid, in-situ readings reported in mg/kg units, offering indicative values suitable for detecting spatial patterns and relative nutrient levels. However, direct electrochemical measurements may deviate from standard laboratory extracts due to soil matrix effects, making them particularly effective for identifying relative highs, lows, and management zones rather than precise absolute concentrations (Hossain et al., 2024).

UAV imaging and electrochemical sensing have progressed largely in parallel. Imaging offers spatial coverage but indirect inference, while electrochemical probes provide specificity without automated deployment. Bridging the two has significant potential. Early prototypes have attempted this integration through UAV–GIS fusion (Chen et al., 2025) and UAV-mounted sampling modules (Kuehl et al., 2025). Ensemble machine learning models, including Random Forests, have consistently outperformed linear methods for nutrient prediction from spatial and soil features (Dash et al., 2025; Hall et al., 2024). Agronomically meaningful nutrient thresholds further improve interpretability and management value (Dey et al., 2024).

This study presents the unique demonstration of a UAV-enabled electrochemical nutrient sensing system capable of automated soil probing, in situ phosphorus (P) measurement and real-time data transmission. Importantly, this work does not attempt to estimate phosphorus from UAV-based remote sensing or spectral imagery. Instead, the UAV is used as an automated deployment and geospatial sampling platform for direct soil sensing, and the contribution of this study lies in integrating in situ electrochemical measurements with spatially informed machine learning to generate continuous phosphorus maps. Field deployments in a Tanzanian flower farm and a Canadian vineyard show that combining direct sensing with geospatial classification reliably identifies agronomic extremes such as high-P zones in Tanzania and low-P zones in British Columbia. Random Forest models trained on nutrient and spatial features outperform baseline classifiers and recover coherent nutrient gradients even with limited sample densities.

By coupling UAV-based automation with direct electrochemical sensing and spatially informed machine learning, this work demonstrates a scalable pathway for generating high-resolution phosphorus maps that overcome the limitations of indirect imaging and sparse laboratory sampling. The results validate the importance of geospatially enriched learning for nutrient mapping and establish a foundation for real-time, data-driven fertilizer management.

2. Methodologies

2.1. Study sites and data collection

Soil sampling was conducted across three study settings. The first site was a vineyard in Abbotsford, British Columbia, Canada (49.11°N, 122.57°W), covering approximately 3.5 ha with a north–south row orientation and established grapevine production. A total of 102 soil samples were collected from the upper 10 cm and georeferenced using a handheld GPS with about five metres accuracy. Nearby samples were aggregated to yield ninety unique spatial locations.

The second site was a flower farm located inside the Aka Khan University in Arusha, Tanzania (3.33°S, 36.61°E), covering about 1.04 ha under intensive floriculture management. Three replicate measurements were collected at each of ninety-nine locations from the top 10 cm of soil, with coordinates recorded using a differential GPS with

approximately one metre accuracy. For comparison with the vineyard dataset, replicate measurements at each location were averaged to produce a single location-level value.

The third study used the full Tanzania dataset without aggregation. All 297 measurements were retained, preserving both the ninety-nine sampling locations and their associated replicates. This design enabled analysis at two spatial scales, capturing between-location structure at the macro level and within-location variability at the micro level, while also supporting a three-class nutrient classification.

Although both sites were sampled using the same UAV-based platform, their soil characteristics differed substantially. The Arusha site is situated on a volcanic loam with near-neutral pH (approximately 6.6–7.4) and regionally low baseline phosphorus and nitrogen levels, with localized enrichment reflecting intensive fertilizer application (Kalonga et al., 2024). In contrast, the British Columbia vineyard occupies a coarse-textured glacial soil with moderate organic matter and predominantly neutral to alkaline pH (> 7.0) (BCTFPG, n.d.), conditions that can reduce phosphorus availability. Potassium levels were generally sufficient, and nitrogen inputs were intentionally limited. These contrasts highlight the need for modeling approaches that remain robust across differing soil chemistries, fertility baselines, and management regimes.

2.2. Spatial feature engineering and autocorrelation analysis

Spatial structure was explicitly incorporated through engineered features derived from GPS coordinates and neighborhood relationships. Positional features included latitude and longitude, polynomial terms (latitude^2, longitude^2, and their interaction) to capture non-linear spatial trends, and Euclidean distances to the field center and to the southwest and northeast corners to account for potential edge effects. Local spatial context was represented using k-nearest neighbors, with k selected empirically based on cross-validation stability across studies. A neighborhood size of k = 5 provided consistently high F1-scores and low-variance under spatially grouped cross-validation in both the two-class and three-class formulations, and was therefore adopted for feature construction.. For each location, neighborhood statistics including the mean, standard deviation, minimum, and maximum phosphorus values were computed, along with the deviation from the neighborhood mean and local sampling density defined by average neighbor distance. This resulted in fourteen location-level features for Studies 1 and 2, from which the five most informative were retained through feature ablation.

Study 3 incorporated repeated measurements per location and therefore included additional reading-level features to represent within-location structure. These included deviation from the location mean, within-location variability, and indicators identifying minimum and maximum readings at each location. In total, nineteen features were generated, and the ten most influential were retained for modeling.

Spatial dependence in the phosphorus data was assessed using Moran's I with k-nearest neighbor spatial weights ($k = 6$), row-standardized and evaluated using 999 permutations under a null hypothesis of spatial randomness. Local Moran's I was used to identify clustered and outlier patterns. For Study 3, spatial structure was examined at two scales: between locations using location means, and within locations using spatial weights defined among repeated readings. Variance partitioning was quantified using the intraclass correlation coefficient, and semi-variograms were used to characterize the relative contributions of micro-scale variability and broader spatial structure through the nugget-to-sill ratio.

2.3. Classification and modeling framework

Phosphorus classification was adapted to each field site because the underlying nutrient distributions were not comparable, as shown in Supplementary Fig. 1. The Tanzania dataset spanned a broad fertility range and supported a three-class formulation representing low, medium, and high phosphorus levels. In contrast, the British Columbia vineyard exhibited a narrower distribution with few high-P observations, making a three-class scheme unstable. As a result, phosphorus values in the BC dataset were reduced to two classes separating deficient from adequate levels. Accordingly, Studies 1 and 2 used a median split to produce two balanced classes, while Study 3 applied tertile thresholds to enable three-class classification in the full Tanzania dataset.

Quantile-based thresholds were selected to produce balanced training classes and to emphasize relative spatial variability within each field. Agronomic phosphorus thresholds vary significantly across soil types, crop systems, and management practices, and consistent site-specific calibration data were not available for the study locations. As a result, the classification scheme is intended to represent relative nutrient zones (e.g., low, medium, high) rather than absolute agronomic sufficiency levels.

Train–test splits were stratified to preserve class balance. Studies 1 and 2 used a 70–30 split at the observation level. For Study 3, splitting was performed at the location level to avoid information leakage, with all repeated readings from the same sampling point assigned to either the training or test set. Feature selection followed a consistent ablation procedure in which features were removed iteratively and model performance was evaluated using F1-score. Five features were retained for the location-level models in Studies 1 and 2, while ten features were selected for the combined location- and reading-level model in Study 3.

Random Forest was selected as the classification model because it captures non-linear relationships, accommodates interactions among spatial and reading-level features, and provides interpretable feature importance measures. Models were implemented using scikit-learn (version 1.6.1) in Python 3.9. Hyperparameters were optimized using RandomizedSearchCV with 5-fold GroupKFold cross-validation, where groups were defined by spatial location to prevent autocorrelation-driven overestimation of performance. The search space included n_estimators {200, 300, 500}, max_depth {15, 20, 25, 30, None}, min_samples_split {2, 5, 10}, min_samples_leaf {1, 2, 4}, max_features {'sqrt', 'log2'}, and class_weight {'balanced', 'balanced_subsample', None}. Weighted F1-score served as the optimization objective. All features were standardized using a StandardScaler fit on the training data only, and no oversampling techniques such as SMOTE were applied.

Model performance was evaluated using accuracy, precision, recall, and F1-score for each class, along with weighted averages to reflect class proportions. Confusion matrices were used to examine classification behavior. A stratified dummy classifier that generated predictions proportional to class frequencies served as a baseline, and improvements were reported as percentage gains in F1-score relative to this reference. Generalization was assessed using five-fold GroupKFold cross-validation, with results summarized as the mean F1-score and corresponding standard deviation.

2.4. Spatial prediction mapping

Spatial predictions were generated on a regular 500 by 500 grid to visualize nutrient patterns across each field. Spatial features for grid points were computed using the same procedures applied to the training data, and neighbor statistics were derived from the training locations. During prediction, neighborhood features were computed exclusively from the training dataset. No ground-truth phosphorus values from prediction locations or test samples were used in feature construction, ensuring that information leakage was strictly avoided.

Three interpolation strategies were compared. To provide meaningful spatial baselines, we compared Random Forest classification against Inverse Distance Weighting (IDW), Ordinary Kriging (OK), and a spatially-interpolated Random Forest. IDW and OK predictions were thresholded at the training-set median to produce binary class labels. All baselines were evaluated on the same held-out test set using weighted F1-score. Supplementary Table 1 provides the weighted F1 results for

comparison of the various methods. The direct Random Forest predictions approach computed grid-point features from the k nearest training neighbors without interpolating phosphorus values. The inverse distance weighting approach used distance-weighted averages of neighboring phosphorus values to estimate neighbor statistics. The hybrid method computed geometric features exactly while applying light weighting for neighbor-based summaries.

Post-processing was applied for visualization. Gaussian smoothing was used to soften predicted surfaces, and field boundaries were delineated using alpha shapes with $\alpha = 0.001$. Predictions outside the sampled region were masked. Maps were rendered using continuous color scales, and sampling locations were overlaid with black circles for training points and white circles for testing points.

2.5. Software and statistical analysis

All analyses were conducted in Python 3.9 using the following packages: pandas (2.2.0) and numpy (1.26.3) for data manipulation; scikit-learn (1.6.1) for machine learning; geopandas (0.14.0), libpysal (4.9.2), and esda (2.5.1) for spatial statistics; matplotlib (3.8.2) for visualization; scipy (1.11.4) for statistical tests; and skgstat (1.0.12) for variogram analysis. Random seed was set to 42 for reproducibility. All code is available in the Supplementary Jupyter notebooks.

Spatial autocorrelation tests employed two-tailed significance at $\alpha = 0.05$. Feature importance values were normalized to sum to 1.0. Model performance comparisons used two-tailed *t*-tests where appropriate.

3. System design and evaluation

3.1. Integrated system overview

The soil nutrient sensing platform developed in this project integrates a quadcopter, robotic soil-probing mechanism and cloud-based data pipeline into a unified system for automated field sampling. Fig. 1 provides an overview of the full UAV based soil sensing workflow, showing how the probe hardware, control electronics and UAV platform work together to collect GPS referenced nutrient measurements and convert them into spatial nutrient maps. The platform is built on a DJI Mavic 3 M (3Multispectral) quadcopter capable of carrying a payload of at least 400 g. AriProbe mass was 320 g (probe 80 g, actuator 120 g, electronics 120 g), within DJI Mavic 3 M limits. A modular attachment, designated AgriProbe, mounts directly to the airframe and houses the mechanical, electrical and sensing components required for in situ nutrient detection. During operation, the UAV handles autonomous navigation between sampling sites, and the AgriProbe module performs soil contact, sensing and data transmission. Unlike aerial spectral methods that infer nutrient conditions indirectly from canopy reflectance, the proposed system operates through direct soil interaction. The UAV lands at each sampling location, and the probe is inserted approximately 10 cm into the soil to obtain in situ electrochemical measurements. Because sensing occurs below the vegetation canopy and within the soil matrix, the presence of vegetation does not affect measurement reliability. This configuration enables continuous, georeferenced sampling across large agricultural fields with minimal operator input.

The system operates as an integrated workflow in which landing, probing, measurement, data acquisition and transmission occur as a coordinated sequence. The mechanical, electrical-control and communication subsystems were designed to interact reliably so that measurements collected at each waypoint are accurately associated with their corresponding GPS coordinates and transferred to the cloud in near real time.

3.2. Mechanical subsystem

The AgriProbe mechanical assembly was designed to balance structural strength, stability and weight. Fig. 2 depicts the hardware layout of the sensing platform, highlighting the soil sensing module, mechanical actuation components and onboard control unit that support in situ nutrient measurements during field operations. The probing mechanism consists of a servo-actuated arm and a linear actuator responsible for inserting the probe to a target depth of approximately 100 mm. The actuator (100 mm stroke, 150 N thrust, 12 V) provides sufficient force for penetration across a range of agricultural soils. Actuator current was monitored qualitatively during probe insertion to detect anomalous resistance events indicative of hard inclusions or compacted zones; insertion depth was maintained consistently across all sampling locations via depth-control stops integrated into the probe assembly. All major structural components, including the arm links, mounting plate and support leg, were 3D-printed in PLA to minimize mass and allow rapid field replacement.

Finite-element and free-body analyses estimated insertion forces near 52 N, which generated upward reaction moments on the UAV. Forces estimated for loamy soils (cone penetrometer data); tested up to

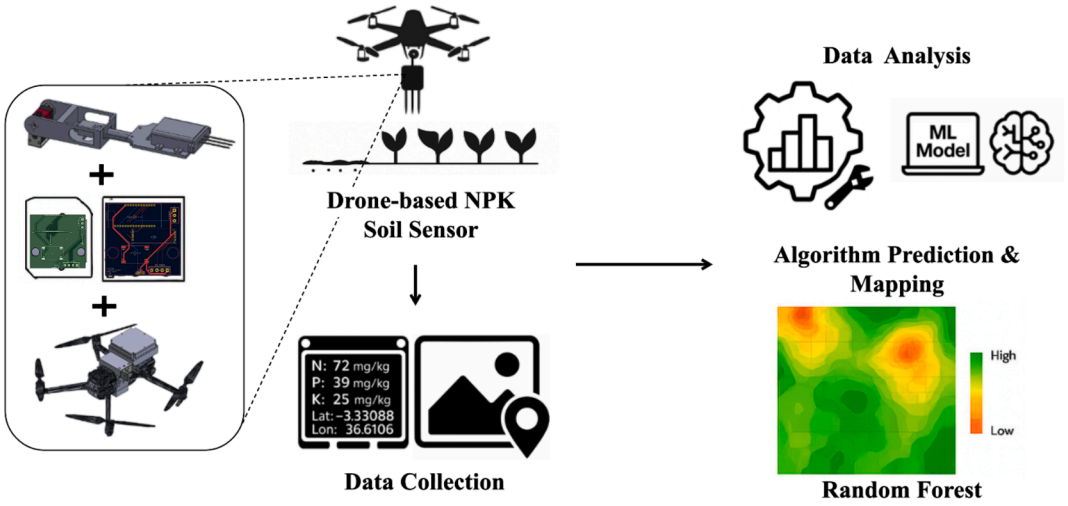


Fig. 1. Overview of drone soil sensing systems. This figure illustrates the workflow of the drone mounted soil sensing platform used to measure N, P, and K concentrations in the field. The system combines custom probe hardware, onboard control electronics, and a UAV capable of automated probe insertion for in situ nutrient sampling. After collecting nutrient readings with GPS coordinates, the data are transmitted for analysis. A machine learning pipeline processes the measurements and generates spatial nutrient maps using a Random Forest classifier, enabling identification of agronomically relevant nutrient patterns across the field.

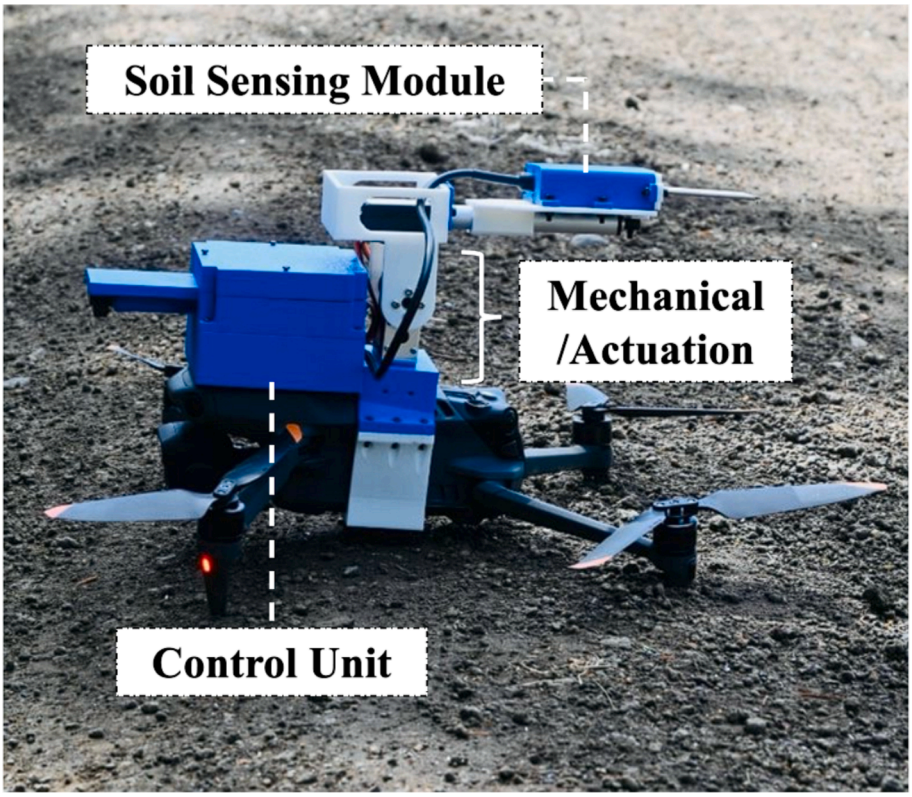


Fig. 2. Hardware setup of the soil sensing system. This figure shows the main hardware components mounted on the UAV platform. The soil sensing module at the top houses the electrochemical NPK probe used for in situ measurements. The mechanical and actuation section is responsible for lowering the probe into the soil during sampling. The control unit positioned near the center of the drone contains the electronics for power distribution, probe communication, and flight coordination. These components form the full sensing assembly that allows the UAV to collect nutrient measurements directly from the field surface.

80 N in sand. To counteract this, a dedicated support leg positioned beneath the actuator transfers loading directly into the ground, allowing the airframe to remain stable during probing. The mounting configuration maintains the center of gravity close to the UAV's neutral axis, contributing to stable landings and takeoff cycles. The use of heat-set inserts and stainless-steel fasteners ensures rigidity under repeated mechanical cycles and moderate wind conditions. A more detailed breakdown of the individual hardware modules integrated into the UAV system can be found in Supplementary Fig. 2. The design evolution of the probing mechanism can be found in Supplementary Fig. 3.

3.3. Electrical and sensing subsystem

The electrical subsystem consolidates sensing, actuation and data handling into a compact control unit centered around an Adafruit ESP32 Feather V2 microcontroller. An RS-485 transceiver (MAX485) facilitates Modbus communication with the NPK sensor, while an H-bridge motor driver and stepped-voltage rails (5 V and 12 V) power the servo and linear actuator. The system runs on a 3.3 V, 3000 mAh lithium-polymer battery with integrated overcurrent protection.

Component layout separates high-current motor circuitry from low-voltage communication lines to reduce electromagnetic interference. Connectors for the servo, actuator, infrared landing sensor and probe allow rapid replacement in the field. The sensing module is a commercial ion-selective electrode probe with independent channels for nitrate, phosphate, and potassium, reporting values in the range 0–2999 mg/kg units and manufacturer-specified $\pm$ 5–10% accuracy (field-validated: RMSE = 4.2 mg/kg vs. lab extracts on n = 20 subsets across sites, confirming relative utility for spatial zoning). These readings provide indicative nutrient levels suitable for relative spatial mapping, as direct in-situ electrochemical measurements can be influenced by soil pH, moisture, ionic interferences, and matrix effects. In this study, we utilized the phosphorus channel to demonstrate relative P distribution and zoning, without site-specific calibration to absolute available P. The IP68-rated body supports operation in moist soils. IP68 supports $>$ 90% RH; deployments limited to $<$ 80% to avoid drift (calibration post-every 10 probes). The Modbus RTU interface provides a reliable digital data stream. This combination enables fast-response nutrient measurements suitable for real-time mapping.

The electrochemical probe used in this study is intended for rapid in situ measurements that capture relative spatial variation rather than precise laboratory-equivalent nutrient concentrations. Accordingly, no site-specific calibration to standardized laboratory extraction methods was performed. This reflects the intended use of the system as a field-deployable sensing tool for identifying spatial patterns, management zones, and relative nutrient differences, rather than replacing conventional laboratory analysis.

Electrochemical measurements in soil are influenced by local conditions such as moisture content, compaction, and ionic background. During field deployment, measurements were therefore taken after ensuring adequate probe–soil contact and allowing a short stabilization period following insertion before recording values. These practices are consistent with the use of portable soil nutrient sensors in precision agriculture, where rapid, approximate measurements are used to support spatial decision-making. As such, the sensing outputs in this study are interpreted as relative indicators of phosphorus distribution across the field.

3.4. Control workflow and data processing

The operational workflow, illustrated in Fig. 3, shows a state-based sequence from landing to cloud visualization. During descent, the infrared proximity sensor detects the ground once clearance drops below 10 cm, after which the controller transitions to the probing state. The

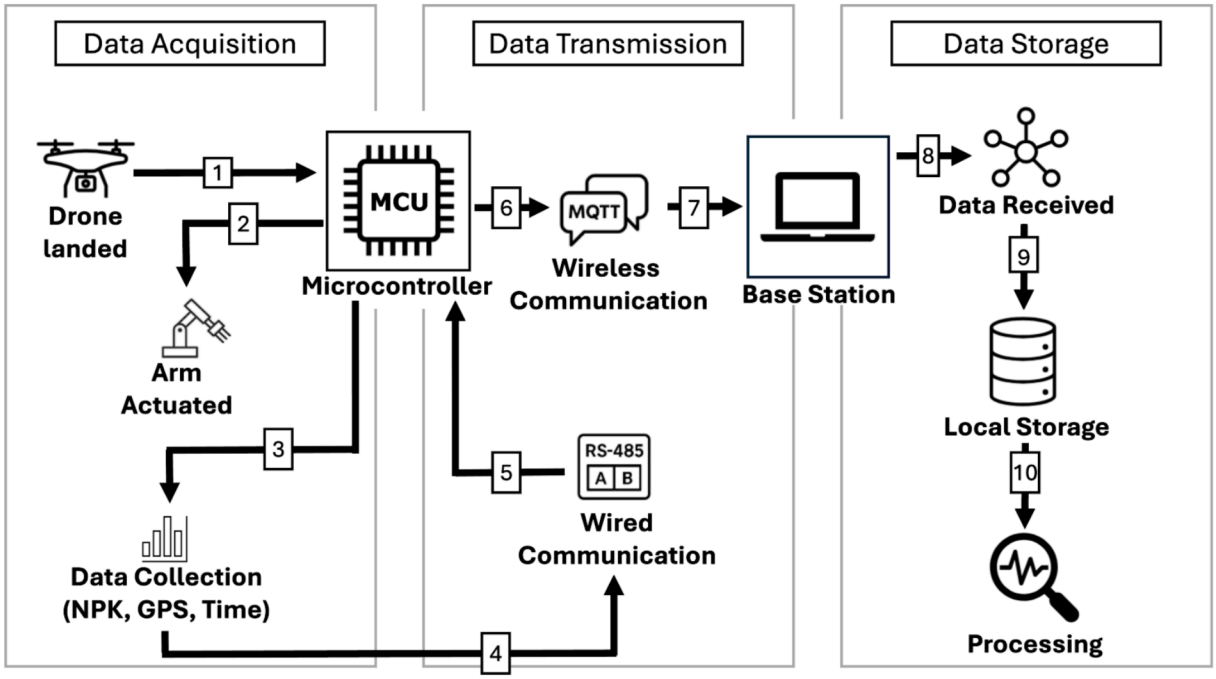


Fig. 3. Control and sensing signal flow. This figure outlines how the system manages data acquisition, transmission, and storage. For data acquisition, the UAV lands and the microcontroller receives the command to actuate the sensing arm. The probe collects nutrient readings along with GPS and time information, and these measurements are sent to the microcontroller for handling. The data transmission section shows how the microcontroller communicates with the base station. Data can be sent through a wired RS 485 link or through wireless transmission using MQTT. The data storage section illustrates what happens once the base station receives the data. The measurements are saved locally and prepared for later processing.

servo rotates the arm into position and the actuator inserts the probe to the target depth while monitoring motor current for potential obstruction. Once insertion is confirmed, nutrient readings are acquired from the probe through the Modbus interface and parsed into mg/kg values by the ESP32.

The controller appends GPS coordinates and timestamps from the UAV telemetry stream and formats the data into a JSON packet. Packets are then transmitted over MQTT to a local or cloud broker. Messages are authenticated and encrypted before being logged into a PostgreSQL database. A Node-RED dashboard provides real-time visualization of nutrient values, UAV position and system status, with automatic CSV generation and SD-card backups for redundancy. After each successful upload, the actuator retracts, the arm returns to its stowed position and the UAV proceeds to the next waypoint. This workflow creates a continuous sensing-to-visualization pipeline for field-scale nutrient mapping.

3.5. Safety, reliability, and usability

Safety mechanisms include hardware interlocks that prevent actuation without landing confirmation, a physical emergency-stop switch that disables all motors and auditory alerts that signal each probing cycle. Firmware-level safeguards include watchdog timers and routines for communication recovery. All electronic assemblies incorporate current limiting, insulation and reinforced solder joints to maintain reliability under vibration.

Field deployments in Tanzania and Canada demonstrated stable operation across contrasting soil textures, moisture levels and environmental conditions. The probing mechanism completed all sampling cycles without mechanical failure or actuator stall, and no data packets were lost during MQTT transmission. The system remained functional through repeated landings and takeoffs, and the support leg maintained UAV stability during probe insertion. These observations indicate that the platform is capable of reliable field operation and can support extended sampling campaigns. Routine maintenance is limited to reprinting worn PLA parts and replacing modular electrical components through JST connectors. The Node-RED dashboard provides operators with real-time system feedback and facilitates immediate data inspection or export for geospatial processing.

3.6. Machine learning framework

Random Forest was selected as the predictive model due to its demonstrated suitability for spatial soil mapping (Cutler et al., 2007; Ghimire, 2023). As an ensemble of decision trees, it can represent nonlinear spatial patterns and interactions among engineered spatial features while providing interpretable feature importance estimates (Breiman, 2001; Louppe, 2015).

Rather than relying solely on coordinates, twelve additional spatial features were engineered and organized into four conceptual categories: (1) polynomial terms (Lat2, Lon2, Lat $\times$ Lon) to capture curved spatial trends; (2) geometric positions such as distances from the field center and corners; (3) neighborhood statistics from the six nearest neighbors (mean, standard deviation, minimum, maximum phosphorus values and local density); and (4) spatial deviation, measuring how much each location differs from its neighbors. This neighborhood-based approach leverages Tobler's First Law of Geography (Tobler, 1970) and has proven effective in precision agriculture applications (Ghimire, 2023; Hengl et al., 2017).

Feature importance analysis revealed that P_diff_from_neighbors dominated with 31–38 percent of total importance across both sites, followed by neighborhood variability and polynomial coordinate terms. Based on this analysis, the top six features were selected for final modeling, reducing dimensionality from fourteen to six while maintaining performance. All features were standardized using zero-mean, unit-variance scaling before training.

3.7. Class imbalance handling

The datasets exhibited substantial class imbalance, particularly for

British Columbia (51% Low, 41% Medium, only 8% High P). This challenge was addressed by defining classification thresholds at the median (binary studies) or tertiles (three-class study), producing balanced class distributions without synthetic oversampling. All samples used in cross-validation and test evaluation correspond to real GPS-referenced field measurements. Synthetic samples were not used at any stage. Spatial cross-validation and test-set evaluation were performed exclusively on real field observations to preserve geospatial realism.

3.8. Hyperparameter optimization and cross-validation

Model hyperparameters were optimized using randomized search with 100 randomly sampled configurations from a search space including n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features, criterion, and bootstrap parameters. Random search can outperform exhaustive grid search when only a subset of hyperparameters strongly influence performance (Bergstra and Bengio, 2012). Each configuration was evaluated using 5-fold StratifiedGroupKFold cross-validation, which maintains both spatial independence, since entire locations stay within folds, and class balance across folds (Roberts et al., 2017). Macro-averaged F1-score was used as the optimization metric, treating all three phosphorus classes equally regardless of their frequency (He and Garcia, 2009). Both sites converged on identical optimal parameters: 500 trees with unlimited depth (max_depth = None), min_samples_split = 10, sqrt feature selection, and entropy splitting criterion.

4. Spatial cross-validation is critical for honest performance estimates

Standard random cross-validation badly overestimates performance when nearby samples are split between training and test sets, allowing models to exploit spatial proximity rather than learning generalizable patterns (Roberts et al., 2017). Rigorous spatial validation was implemented by splitting unique locations, rather than individual samples, into 70% training and 30% testing, which ensured complete spatial separation. All performance metrics reported in Results were calculated from these spatially independent test locations.

4.1. Model evaluation

Performance was assessed using precision, defined as the proportion of correct predictions among predicted positives for each class, recall, defined as the proportion of actual instances correctly identified, F1-score, defined as the harmonic mean of precision and recall, and confusion matrices. Macro-averaged F1-score served as the primary metric because it treats all classes equally and is not inflated by majority class predictions, which is important for imbalanced datasets (Powers and Atyabi, 2012). Comparison was made against a naive baseline (DummyClassifier predicting the most frequent class), which achieved macro F1-scores of only 0.035–0.143, demonstrating the substantial value added by the spatial modeling approach.

5. Spatial predictions and visualization

To generate continuous maps, 500 × 500 prediction grids (250,000 points) were created spanning each study area. Neighborhood features at grid points were computed by identifying the six nearest training locations and calculating their statistics, assuming that spatial autocorrelation allows reasonable approximation from nearby samples (Sekulić et al., 2020). Random Forest class probabilities were converted into a continuous Phosphorus Class Index (0 = Low, 1 = Medium, 2 = High) through weighted scoring:

$$\text{Index} = P(\text{Low}) \times 0 + P(\text{Medium}) \times 1 + P(\text{High}) \times 2.$$

This approach preserves prediction of uncertainty information that would be discarded by hard classification.

To quantify spatial prediction uncertainty, Shannon entropy was computed from the Random Forest class probabilities at each grid point, $H = -\Sigma_k p_k \log(p_k)$, where p_k represents the predicted probability of class k. High entropy values indicate uncertain predictions, typically near class boundaries, while low entropy reflects high model confidence. The resulting uncertainty surface is shown alongside the class prediction maps in Fig. 4. As shown in Fig. 4a, in Study 1 uncertainty is concentrated along the central transition zone separating low- and high-phosphorus regions, while more homogeneous areas exhibit consistently low entropy. Fig. 4b shows Study 2, where uncertainty aligns with the primary spatial gradient, with increased confidence observed within clearly defined nutrient zones. In Study 3, shown in Fig. 4c, higher entropy values appear predominantly in intermediate regions, reflecting increased ambiguity associated with the three-class formulation.

Gaussian smoothing ($\sigma = 2$ grid cells, approximately 20–30 m) was applied solely to the rasterized prediction surface for cartographic visualization; all accuracy metrics were calculated on unsmoothed predictions. The smoothing kernel was selected to be substantially smaller than the empirical variogram range (≈ 85 m), ensuring that no spatial structure beyond the native prediction resolution is introduced. Alpha shape boundary delineation restricted predictions to the actual sampling domain, with $\alpha = 0.001$ selected to balance tight boundaries against exclusion of valid interpolation areas (Edelsbrunner et al., 1983). Grid cells outside this boundary received NaN values, which appear as white space in maps to clearly distinguish interpolated predictions from unreliable extrapolations.

All analyses were conducted using Python 3.9 + with scikit-learn (v1.6 +), NumPy, pandas, SciPy, alphashape, Shapely, and Matplotlib. Fixed random seeds (random_state = 42) were used throughout all stochastic procedures to ensure complete reproducibility. Complete analysis scripts are provided in the Supplementary Materials.

5.1. Algorithm workflow

The algorithmic workflow follows a structured sequence beginning with data preparation, in which field-sampled soil measurements collected by the AgriProbe–SkySoil system are paired with their corresponding GPS coordinates. Fig. 5 summarizes the mapping algorithm workflow from data collection through preprocessing, binning, model training and visualization. Continuous nitrate, phosphate, and potassium values are then discretized into nutrient classes using quantile-based thresholds. The model-training stage uses 70% of these labeled samples to fit a Random Forest classifier, with latitude and longitude serving as the primary predictive features. Hyperparameters such as *n_estimators*, *max_depth*, and *min_samples_split* are tuned through cross-validation to achieve an optimal balance between bias and variance. Once trained, the model is applied to a fine-resolution spatial mesh generated over the study's geographic bounding box, creating a dense grid of virtual sampling points. For each point in this mesh, the classifier returns predicted nutrient-class probabilities, which are then visualized through heatmaps or contour layers to form probabilistic nutrient-distribution maps. Model performance is evaluated using a held-out test subset, drawing on metrics such as accuracy, F1-score, and confusion matrices, while spatial cross-validation is used to mitigate location-based bias. Overall, this sensing-to-processing-to-modeling workflow transforms localized UAV-acquired soil measurements into continuous field-scale nutrient maps. By combining real-world sensor data with statistical learning, the system enables scalable inference of nutrient conditions across areas where no physical samples were collected.

6. Results and discussion

Field deployments in Tanzania and British Columbia demonstrated that the UAV-mounted sensing system could reliably execute repeated

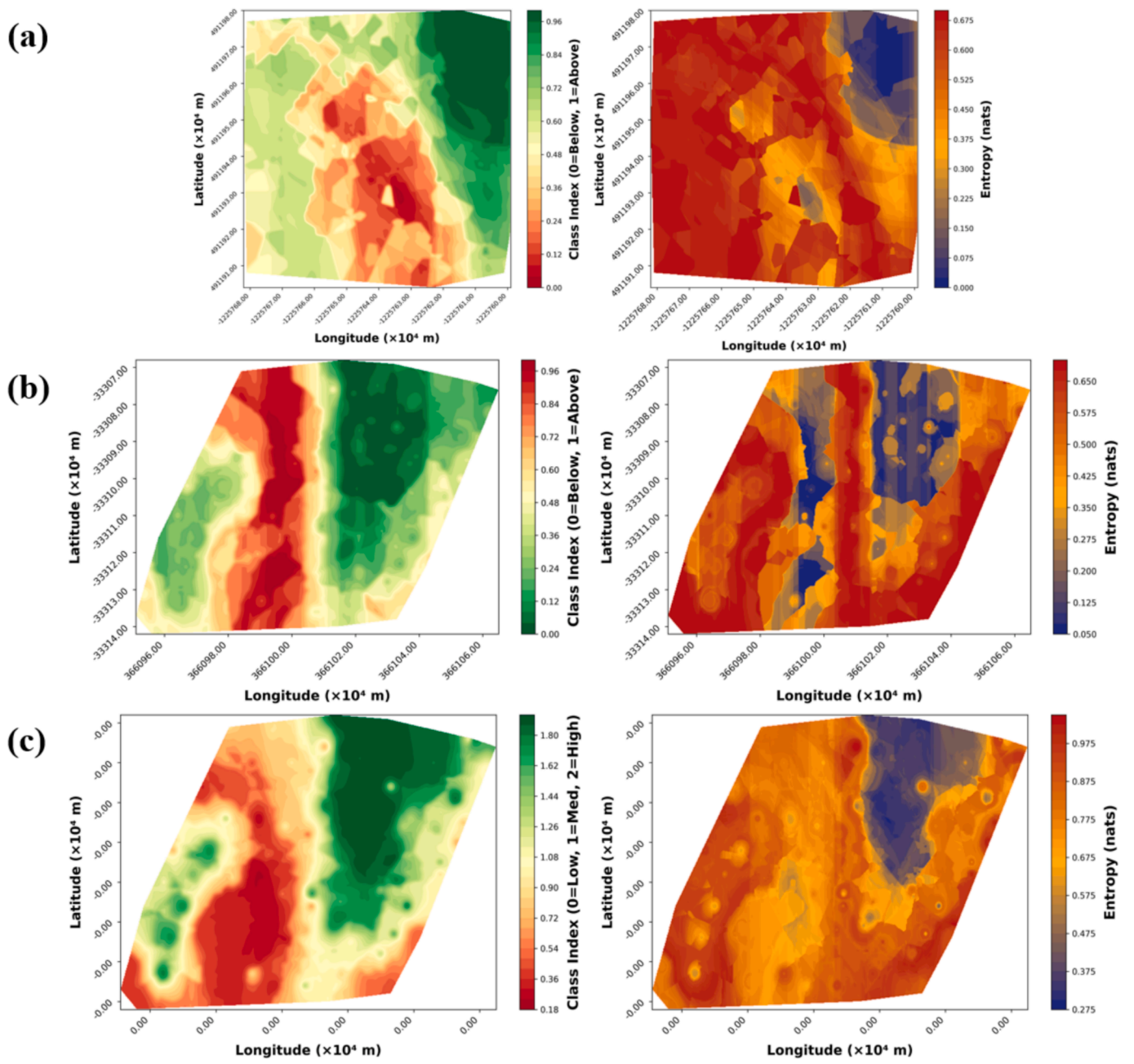


Fig. 4. Spatial phosphorus classification and prediction uncertainty across study sites. (a) British Columbia vineyard (Study 1). Left: predicted phosphorus class index (0 = below median, 1 = above median) generated using the spatial Random Forest model, showing a west–east nutrient gradient. Right: corresponding prediction uncertainty map quantified using Shannon entropy, where higher values indicate lower model confidence, primarily near class transition zones. (b) Tanzania flower farm (Study 2). Left: predicted two-class phosphorus surface showing a strong spatial gradient across the field. Right: entropy-based uncertainty map highlighting increased confidence within homogeneous regions and higher uncertainty along class boundaries. (c) Tanzania full dataset with three-class formulation (Study 3). Left: predicted phosphorus class index (0 = low, 1 = medium, 2 = high) incorporating within-location variability. Right: corresponding uncertainty map, with elevated entropy in intermediate regions reflecting increased ambiguity between adjacent classes.

autonomous soil sampling cycles under real agricultural conditions. Across both sites, the system consistently completed the full operational sequence of landing, probe deployment, soil insertion, nutrient measurement, probe retraction, and data transmission without critical failures. The linear actuator enabled vertical probe insertion to a depth of approximately 10 cm, targeting the active root-zone soil relevant for nutrient assessment. Post-deployment evaluations indicated that the redesigned single-axis actuation mechanism reduced lateral forces during insertion compared to earlier prototypes, improving ground stability and preventing upward reaction forces on the UAV during probing. Field trials suggested stable nutrient readings across multiple insertion cycles and soil textures, indicating consistent probe–soil contact under both tilled and moderately compacted conditions.

Operational timing and flight constraints further illustrate the system's feasibility for field-scale use. Individual sampling cycles required approximately 60–90 s from landing to takeoff, including probe insertion, sensing, and data transmission. Total flight endurance with the sensing payload was estimated at 18–22 min per battery. Under these conditions, a single flight can support approximately 12–20 sampling events depending on waypoint spacing and operational margins, indicating that the current platform is best interpreted as a prototype for targeted high-resolution sampling rather than full-field coverage. Power

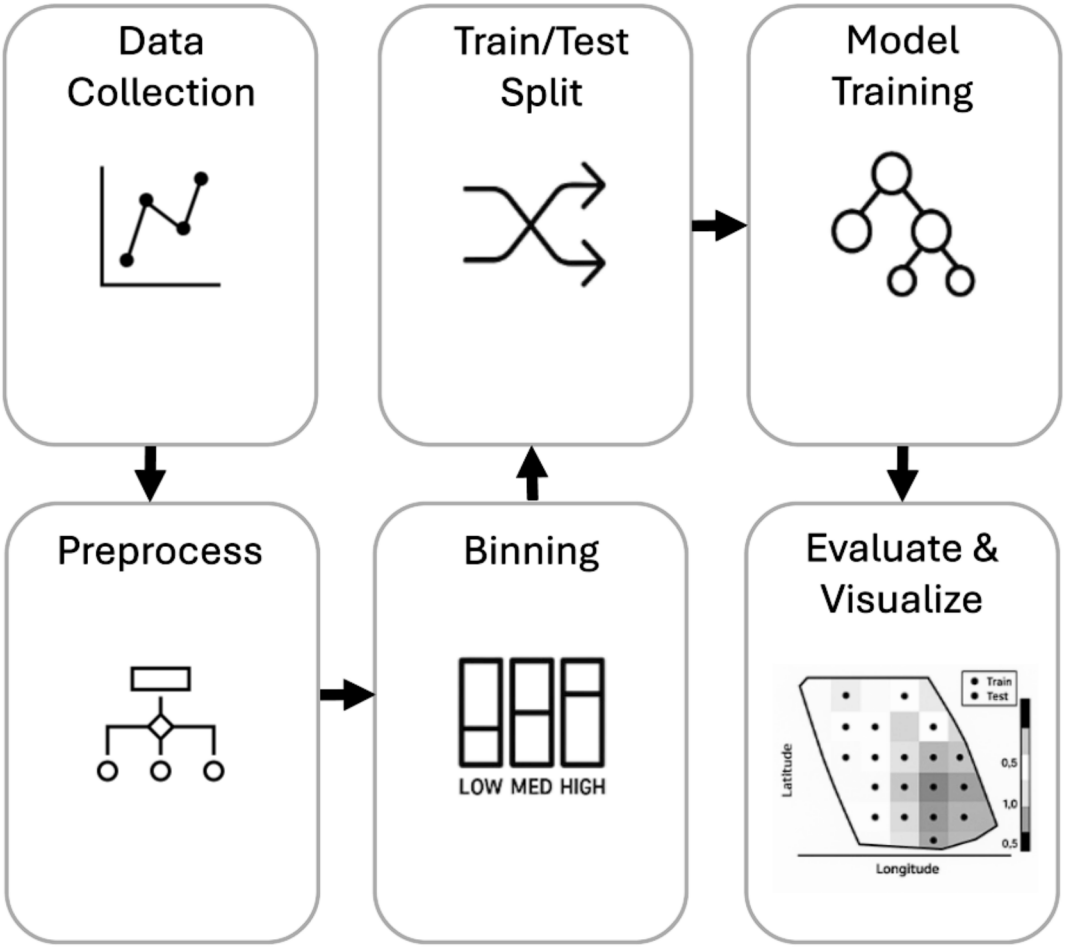


Fig. 5. Mapping algorithm workflow. This figure presents the main steps used to generate the nutrient maps. The workflow begins with raw data collection from field measurements. The data are then preprocessed, which includes cleaning and organizing the inputs so they are ready for analysis. A binning step is applied to assign each nutrient reading to low, medium, or high classes. After binning, the data are split into training and testing sets. The machine learning model is trained using the prepared data, and the final panel shows how the outputs are evaluated and visualized to produce nutrient distribution maps.

consumption remained within the payload limits of a DJI Mavic 3–class platform, with total system mass maintained below 600 g, preserving flight stability and autonomous navigation performance. With further refinement, the additional weight of the hardware can be reduced further, enabling less power consumption and longer flight time, while future scalability may be achieved through reduced sensing-cycle duration, higher-payload UAV platforms, or coordinated multi-UAV deployment. No data packets were observed to be lost during MQTT transmission in the reported field trials, and repeated landing and probing cycles did not produce observable degradation in mechanical or electrical performance over the duration of the deployments. Early testing in Tanzania identified soil compaction and surface dryness as limiting factors, motivating the addition of a ground-contact support leg and improved actuator alignment. Subsequent validation in British Columbia confirmed that these refinements enabled reliable probe insertion without actuator stalling or sensor misalignment. Together, these results show that the platform can sustain repeated autonomous soil measurements with consistent insertion depth, stable sensing behavior, and a robust sensing-to-transmission workflow, providing a validated physical foundation for the spatial modeling results presented below.The Random Forest models across the three studies produced phosphorus classifications that were spatially coherent, agronomically plausible and stable under grouped cross-validation. Each study provided a different nutrient distribution and sampling structure, and the results show how the model adapted to these differences while maintaining consistent performance.

To assess sensitivity to sample density, the Random Forest model was retrained on subsampled training sets of 50% and 30% of the original samples, preserving location-based stratification. For Study 1 in the BC vineyard, as shown in Fig. 6a, weighted F1-scores of 0.849 (30%), 0.774 (50%), and 0.851 (100%) indicate strong robustness to sample reduction, with performance remaining above 0.77 across all density levels. For Study 2 in the Tanzania site, as shown in Fig. 6b, F1-scores of 0.766 (30%), 0.833 (50%), and 0.900 (100%) demonstrate a consistent improvement with increasing sample density, confirming expected behavior. For Study 3 in the Tanzania site, as shown in Fig. 6c, F1-scores of 0.876 (30%), 0.876 (50%), and 0.893 (100%) indicate minimal performance degradation under reduced sampling, further supporting model robustness across varying data densities. These results suggest that the framework remains viable even under constrained sampling budgets, demonstrating consistent robustness across multiple study configurations, which is particularly relevant for smallholder farming contexts where extensive field campaigns may not be feasible.

The nutrient measurements in Table 1a reveal that the British Columbia vineyard has a compressed phosphorus range, with early sample values between 3 mg/kg and 16 mg/kg and similar compression in nitrogen and potassium. The close clustering of latitudes and longitudes indicates that the dataset reflects a compact field with limited chemical variability. These characteristics are visible in the point distribution in Fig. 7a. Despite this narrow spread, the model reconstructs a clear nutrient gradient in Fig. 7b, identifying low phosphorus levels across the western and central parts of the vineyard and higher levels

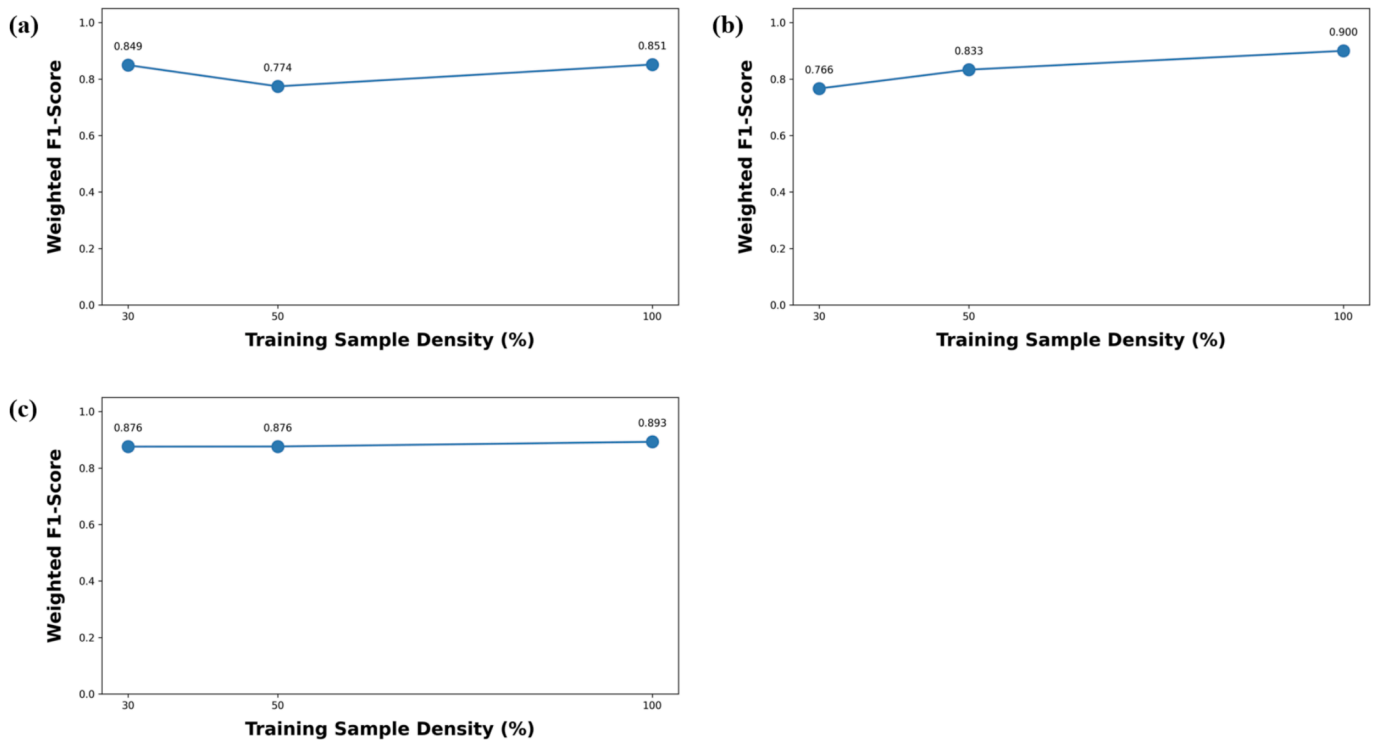


Fig. 6. Model performance sensitivity to training sample density across study sites. (a) British Columbia vineyard (Study 1). Weighted F1-score as a function of training sample density (30%, 50%, 100%), showing stable performance despite reduced sampling, with minimal degradation between 30% (0.849) and 100% (0.851) and a moderate dip at 50% (0.774). (b) Tanzania flower farm (Study 2). Weighted F1-score increases consistently with training density, from 0.766 at 30% to 0.900 at full sampling, indicating stronger dependence on sample availability under higher spatial variability. (c) Tanzania full dataset with three-class formulation (Study 3). Performance remains highly stable across all sampling densities, with F1-scores between 0.876 and 0.893, demonstrating robustness of the model when incorporating within-location variability. All results are reported using spatially grouped validation to preserve independence between training and test locations.

Table 1
Nutrient measurements and model performance for Study 1. Panel (a) shows a sample of the nutrient measurements collected for Study 1, including nitrogen, phosphorus, and potassium values along with the corresponding GPS coordinates. Panel (b) lists the performance metrics for the Study 1 model, including accuracy, baseline F1 score, macro F1 score, weighted F1 score, and the mean and standard deviation of the cross-validation accuracy.

(a) Study 1 data sample

Point Number	N (mg/kg)	P (mg/kg)	K (mg/kg)	Latitude	Longitude
1	3	4	9	49.11908	−122.57601
2	12	16	33	49.11932	−122.57599
3	8	11	23	49.11938	−122.57601
4	2	3	6	49.11955	−122.57602
5	2	4	9	49.11966	−122.57602

(b) Study 1 performance metrics

Metric	Value
Accuracy (Test set)	0.815
Baseline F1-score	0.515
Macro F1-score	0.814
Weighted F1-score	0.813
Mean CV Accuracy (GroupKFold)	0.911
Std CV Accuracy (GroupKFold)	0.075

toward the east. The per-class metrics in Fig. 7c show that the two classes are separated with F1-scores of 0.86 for low and 0.85 for high, and the cross-validation curve in Fig. 7d is consistent with the mean GroupKFold accuracy of 0.911 shown in Table 1b. The Supplementary classification comparison in Supplementary Table 2 echoes these results, with a macro F1-score of 0.81 for Study 1. Supplementary Table 3 places these findings in a broader context, showing that fold accuracies for Study 1 span from 0.7778 to 1.0000, reflecting stable but variable performance under different group splits.

The feature i mportance results in Table 4 explain how the model compensates for the narrow phosphorus range in Study 1. The most influential feature, P_diff_from_neighbors, has a value of 0.3682, indicating that local spatial contrast is the primary signal used to distinguish classes. Neighbor_std_P and the corner distance measures dist_to_sw and dist_to_ne contribute additional spatial information, while longitude captures the east–west gradient visible in Fig. 7b. These results show that the model relies on spatial descriptors rather than raw nutrient magnitude when field variation is limited.

The Tanzania flower farm dataset in Study 2 presents a broader nutrient range, as shown in Table 2a, where phosphorus spans from 17 mg/kg to 28 mg/kg among the first sample points. This wider variation supports stronger spatial separability, which is reflected in Fig. 8a and in the predicted surface in Fig. 8b. The model achieves balanced and slightly higher per-class performance than in Study 1, with F1-scores of 0.90 for both low and high phosphorus levels. The cross-validation trend in Fig. 8d and the performance metrics in Table 2b show that the model remains stable, with a mean GroupKFold accuracy of 0.889 and a standard deviation of 0.049. Supplementary Table 2 reinforces this improvement, with macro and weighted F1-scores of 0.87. Supplementary Table 3 shows a narrower accuracy range for Study 2 than Study 1, from 0.8000 to 0.9504, which reflects the stronger nutrient gradients that simplify the classification task.

The feature importance rankings in Table 5 align with these patterns. Although P_diff_from_neighbors again ranks first with an importance of 0.3738, longitude and lon_squared also carry substantial weight, suggesting that the model makes greater use of absolute spatial position when clearer nutrient gradients are present. The neighborhood features neighbor_max_P and neighbor_mean_P provide additional local context, supporting the smooth spatial trend seen in Fig. 8b.

Study 3 introduces repeated measurements at each location, and

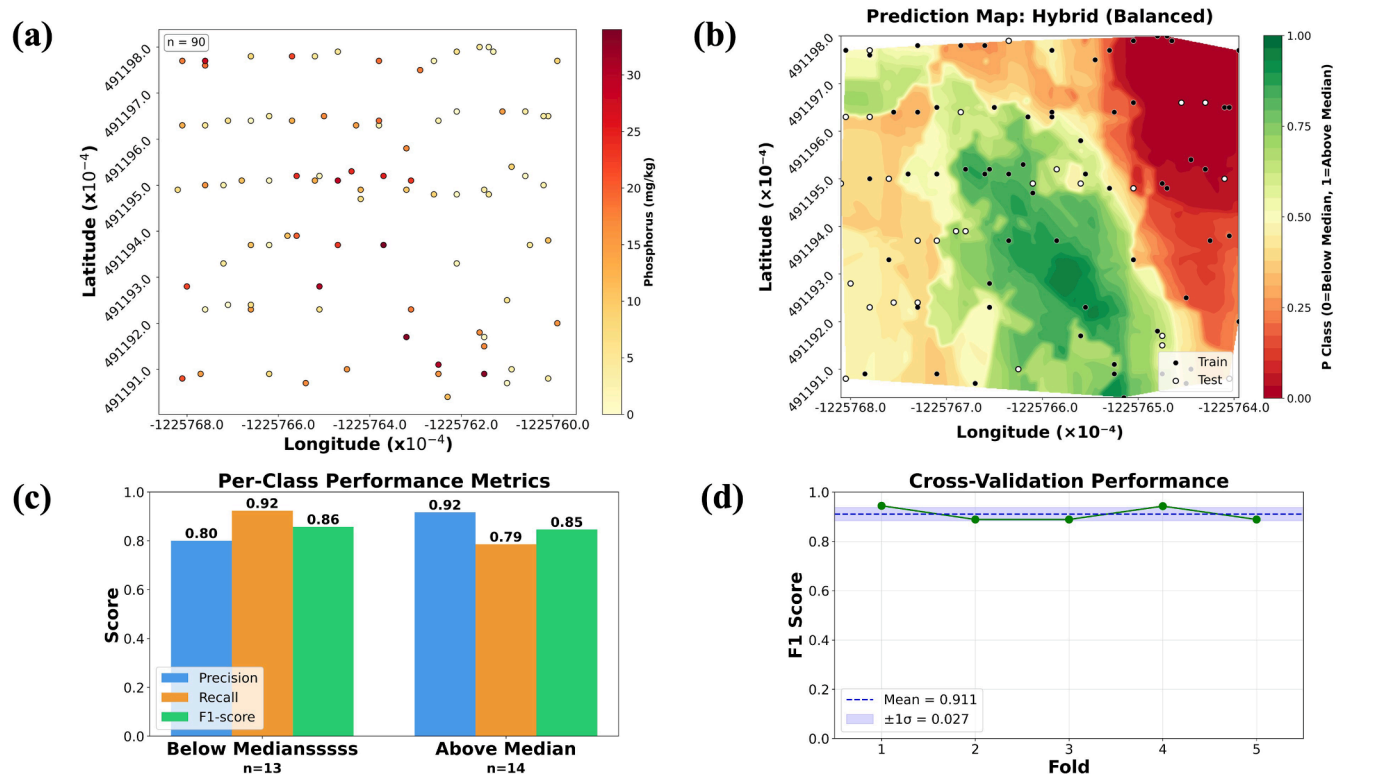


Fig. 7. Algorithm performance Study 1 in British Columbia. (a) Spatial distribution of sampled soil phosphorus measurements across the vineyard, showing the narrow nutrient range and compact field geometry. (b) Predicted phosphorus classification surface generated using the spatial Random Forest model, illustrating a west–east nutrient gradient despite limited absolute variability. (c) Classification performance on the spatially independent test set, reported as precision, recall, and F1-score for the low- and high-phosphorus classes. (d) GroupKFold cross-validation performance showing model stability across spatial folds. Black circles indicate training locations and white circles indicate testing locations. Predictions are masked outside the sampled domain.

Table 2
Nutrient measurements and model performance for Study 2. Panel (a) presents a sample of the nutrient measurements used in Study 2, with nitrogen, phosphorus, and potassium values shown together with latitude and longitude. Panel (b) summarizes the model performance metrics for Study 2, including accuracy, baseline F1 score, macro and weighted F1 scores, and the mean and standard deviation of the cross-validation accuracy.

a) Study 2 data sample					
Point Number	**N (mg/kg)**	**P (mg/kg)**	**K (mg/kg)**	**Latitude**	**Longitude**
1	19	28	58	−3.33076	36.61065
2	12	18	36	−3.33082	36.61062
3	18	26	53	−3.33082	36.61062
4	16	23	47	−3.33088	36.61059
5	12	17	35	−3.33095	36.61055

(b) Study 2 performance metrics	
Metric	**Value**
Accuracy (Test set)	0.887
Baseline F1-score	0.531
Macro F1-score	0.867
Weighted F1-score	0.867
Mean CV Accuracy (GroupKFold)	0.889
Std CV Accuracy (GroupKFold)	0.049

Table 3
Nutrient measurements and model performance for Study 3. Panel (a) provides a sample of the full dataset used in Study 3, including all replicate measurements for nitrogen, phosphorus, and potassium together with their GPS coordinates. Panel (b) reports the performance metrics for the Study 3 model, including accuracy, baseline F1 score, macro and weighted F1 scores, and the mean and standard deviation of cross-validation accuracy.

(a) Study 3 data sample					
Point Number	**N (mg/kg)**	**P (mg/kg)**	**K (mg/kg)**	**Latitude**	**Longitude**
1	14	20	42	−3.33076	36.61065
1	23	33	67	−3.33076	36.61065
1	21	30	65	−3.33076	36.61065
2	10	15	31	−3.33082	36.61062
2	14	20	40	−3.33082	36.61062
2	13	19	38	−3.33082	36.61062
3	13	19	41	−3.33082	36.61062
3	17	25	51	−3.33082	36.61062
3	24	33	67	−3.33082	36.61062
4	14	20	41	−3.33088	36.61059
4	17	24	50	−3.33088	36.61059
4	17	24	49	−3.33088	36.61059

(b) Study 3 performance metrics	
Metric	**Value**
Accuracy (Test set)	0.893
Baseline F1-score	0.453
Macro F1-score	0.880
Weighted F1-score	0.895
Mean CV Accuracy (GroupKFold)	0.921
Std CV Accuracy (GroupKFold)	0.030

Table 3a shows that these readings can differ substantially. Fig. 9a shows the complete phosphorus point dataset collected in Tanzania. The model nonetheless reconstructs a coherent phosphorus surface in Fig. 9b that extends the structure observed in Study 2. The classification report in Fig. 9c shows strong performance for the low and high classes, with F1-scores of 0.92 and 0.95, while the medium class achieves an F1-score of 0.76. This difficulty is expected because intermediate nutrient levels often overlap with their neighboring classes both numerically and

Table 4
Feature importance for Study 1. This table lists the features used in the Study 1 model along with their descriptions and their corresponding importance values. The features include local phosphorus contrast, neighbour based variability, distances to reference corners of the field, and the longitude of each point.

Feature	Description	Importance
P_diff_from_neighbors	*Point P minus neighbour mean (local contrast)*	0.3682
neighbor_std_P	*Variability of neighbours' P values*	0.0751
dist_to_sw	*Distance to field's southwest corner*	0.0727
dist_to_ne	*Distance to field's northeast corner*	0.0687
longitude	*East–west position of the point*	0.0677

spatially. The high stability of the model in Study 3 is reflected in Fig. 9d and in Table 3b, where the mean GroupKFold accuracy is 0.921 with a standard deviation of 0.030. Supplementary Table 2 includes a broader comparison of the three-class results, and Supplementary Table 3 shows that Study 3 has the tightest accuracy range of all studies, from 0.8889 to 0.9646.

The feature importance results in Table 6 show how the model integrates both multi-reading and spatial information in Study 3. Location_mean_P, with an importance of 0.2715, is the most influential feature because it captures the central tendency of each location's phosphorus condition. P_diff_from_neighbors and P_deviation_from_location_mean capture spatial contrast and within-location variation, while is_location_min and is_location_max help differentiate intermediate values from their extremes. Spatial position remains relevant through features such as dist_to_ne, longitude and lon_squared, and neighborhood averages continue to provide useful contextual information. This combination highlights the model's flexibility when managing a more complex label structure.

The confusion matrices in Fig. 10 summarize how these classification patterns manifest across the three studies. In Study 1, compressed nutrient values result in symmetric errors between the two classes. Study 2 shows fewer misclassifications due to clearer nutrient gradients. Study 3 exhibits most errors between adjacent classes, which aligns with the medium class performance in Fig. 9c and the distributional variability evident in Table 3a. Across all studies, extreme classes remain well separated, and the model rarely confuses the lowest and highest nutrient levels.

The outputs depicted in Fig. 11 show how the predictions from all three studies can be incorporated into a decision support interface. The dashboard allows the user to toggle between crop types, nutrients and predicted surfaces, and the point map over satellite imagery helps confirm that model outputs align with recognizable field features. The soft classification map and the three-zone fertilizer recommendation map translate model predictions into practical guidance, where each spatial zone corresponds to a recommended fertilizer intensity.

Table 5
Feature importance for Study 2. This table presents the features used in the Study 2 model, with descriptions and importance values. The features include local phosphorus contrast, longitude, a squared longitude term, the maximum phosphorus value among neighbouring points, and the mean phosphorus value of neighbouring points.

Feature	Description	Importance
P_diff_from_neighbors	*Point P minus neighbour mean (local contrast)*	0.3738
longitude	*East–west position of the point*	0.0769
lon_squared	*Curvature in east–west spatial trend*	0.0746
neighbor_max_P	*Highest neighbour P value*	0.0711
neighbor_mean_P	*Average P of nearest neighbours*	0.0676

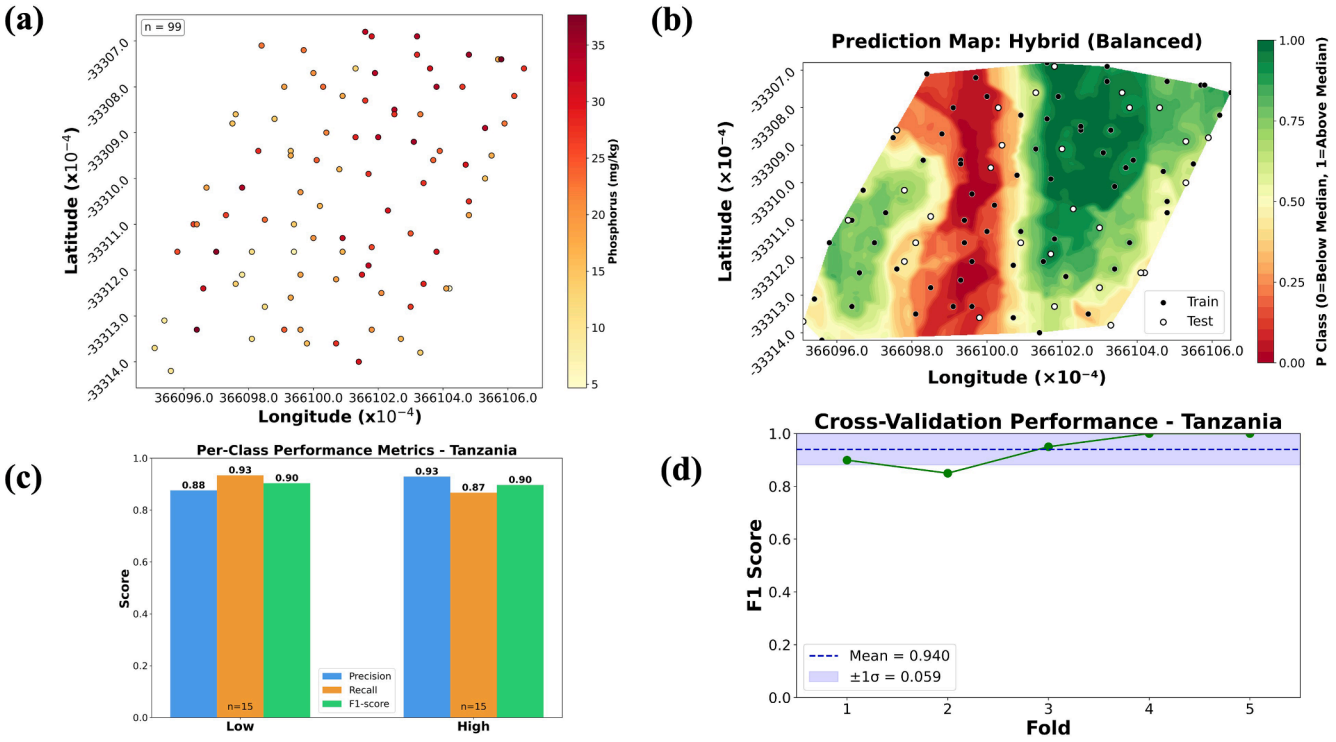


Fig. 8. Algorithm performance Study 2 in Tanzania. (a) Spatial distribution of location-averaged soil phosphorus measurements across the study area, exhibiting a broader nutrient range than the British Columbia site. (b) Predicted phosphorus classification surface obtained from the spatial Random Forest model, showing clear and coherent nutrient gradients across the field. (c) Classification metrics for the low- and high-phosphorus classes evaluated on spatially independent test locations. (d) GroupKFold cross-validation results demonstrating consistent model performance across spatial splits.Black circles represent training points and white circles represent testing points. Predicted values are displayed only within the convex sampling boundary.

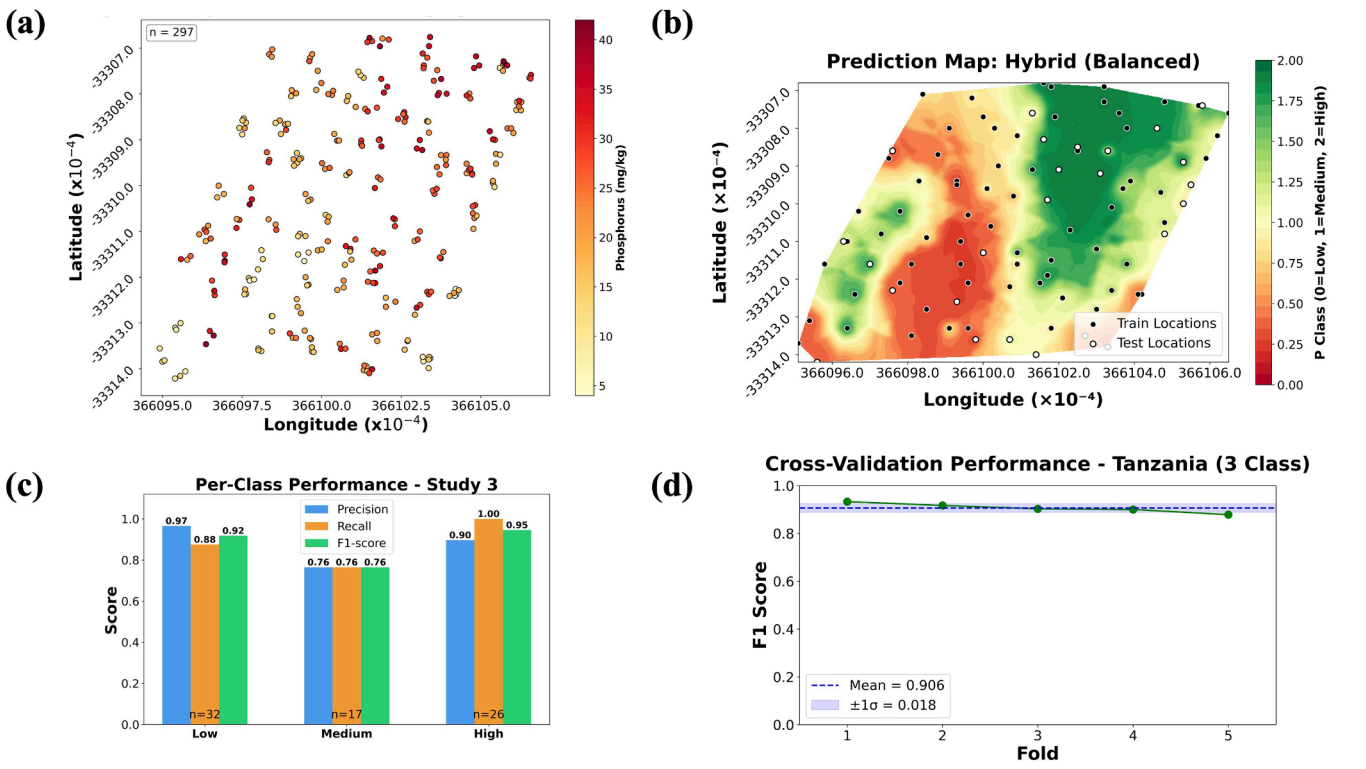


Fig. 9. Algorithm performance Study 3 in Tanzania. (a) Complete spatial distribution of all individual phosphorus measurements, including repeated readings collected at each sampling location. (b) Predicted three-class phosphorus surface produced by the Random Forest model incorporating both spatial and within-location features. (c) Classification report showing precision, recall, and F1-score for low, medium, and high phosphorus classes, highlighting increased uncertainty in intermediate nutrient levels. (d) GroupKFold cross-validation performance illustrating strong stability despite increased classification complexity. Black markers indicate training observations and white markers indicate test observations. Spatial predictions are constrained to the sampled region.

Table 6
Feature importance for Study 3. This table lists the features used in the Study 3 model, along with their definitions and corresponding importance values. The features include location level phosphorus summaries, local contrast features, indicators of whether a reading is the minimum or maximum at its location, distances to field reference corners, longitude-based terms, and neighbour based phosphorus values.

Feature	Definition	Importance
location_mean_P	*Mean P from the 3 readings at that location*	0.2715
P_diff_from_neighbors	*Point P minus neighbour mean (local contrast)*	0.1411
P_deviation_from_location_mean	*Reading's difference from its location mean*	0.1159
is_location_min	*1 if this reading is the lowest at its location*	0.0518
dist_to_ne	*Distance to field's northeast corner*	0.0501
longitude	*East–west position of the point*	0.0385
lon_squared	*Curvature in east–west spatial trend*	0.0339
neighbor_mean_P	*Average P of nearest neighbours*	0.0333
is_location_max	*1 if this reading is the highest at its location*	0.0310
dist_to_sw	*Distance to field's southwest corner*	0.0289

Combined with the stability metrics in Supplementary Table 3 and the consistent classification performance in Supplementary Table 2, the results indicate that the modeling approach can support real-time nutrient monitoring and variable-rate fertilizer decisions. The three studies show how the Random Forest models responded to different field conditions and label structures, and the comparison helps explain why performance varied across sites and nutrient categories. The BC vineyard in Study 1 had a narrow phosphorus range, which is clear from the sample values in Table 1a and from the point distribution in Fig. 7a. This made the classification problem more subtle because the differences between locations were small. Even so, the model was able to recover a consistent gradient in Fig. 7b and reach stable performance across folds in Table 1b and Fig. 7d. When looking at these results next to the Tanzania field in Study 2, the contrast becomes clearer. Table 2a shows much wider nutrient variation in Tanzania, and the spatial pattern in Fig. 8a forms a stronger gradient than the one in the BC field. This helped the model separate the two classes more confidently, which is visible in the higher F1-scores in Fig. 8c and the improved accuracy in Table 2b. The confusion matrices in Fig. 10a and Fig. 10b show this difference well. Study 1 has more back and forth errors between the two classes, while Study 2 reduces these errors and places more observations correctly. Supplementary Table 2 supports this comparison, since the macro averages in Tanzania are consistently higher than those in BC.

The feature importance rankings help explain why performance improved from Study 1 to Study 2. In both tables, P_diff_from_neighbors is the top feature, which shows that local contrast was important at both sites. However, the supporting features tell a different story. In the BC vineyard, features like neighbor_std_P and the corner distances in Table 4 mattered more, suggesting the model needed extra spatial cues to pick up the weaker gradient. In Tanzania, longitude and lon_squared in Table 5 ranked higher, which matches the smoother east–west trend seen in Fig. 8b. The stronger nutrient differences in Tanzania meant the model could rely more on overall field position and neighbourhood summaries and less on subtle contrasts.

Comparing Study 2 and Study 3 shows how the model behaves when the label structure becomes more detailed within the same field. Study 2

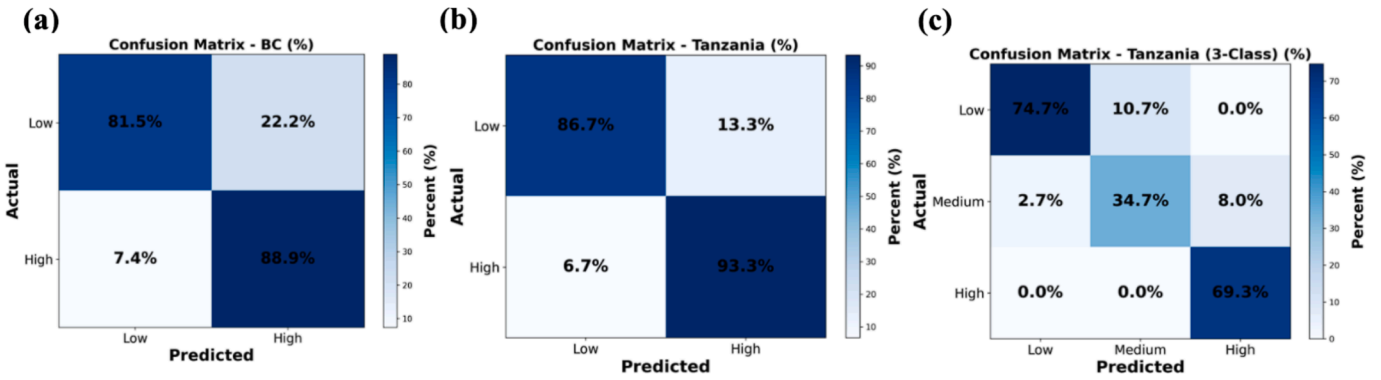


Fig. 10. Confusion matrices of the three studies. (a) shows the confusion matrix for the two-class classification model applied to the BC dataset, with percentages indicating the proportion of predictions in each cell. (b) presents the confusion matrix for the two-class model applied to the aggregated Tanzania dataset. (c) displays the confusion matrix for the three-class model using the full Tanzania dataset, showing the distribution of predictions across the low, medium, and high classes.

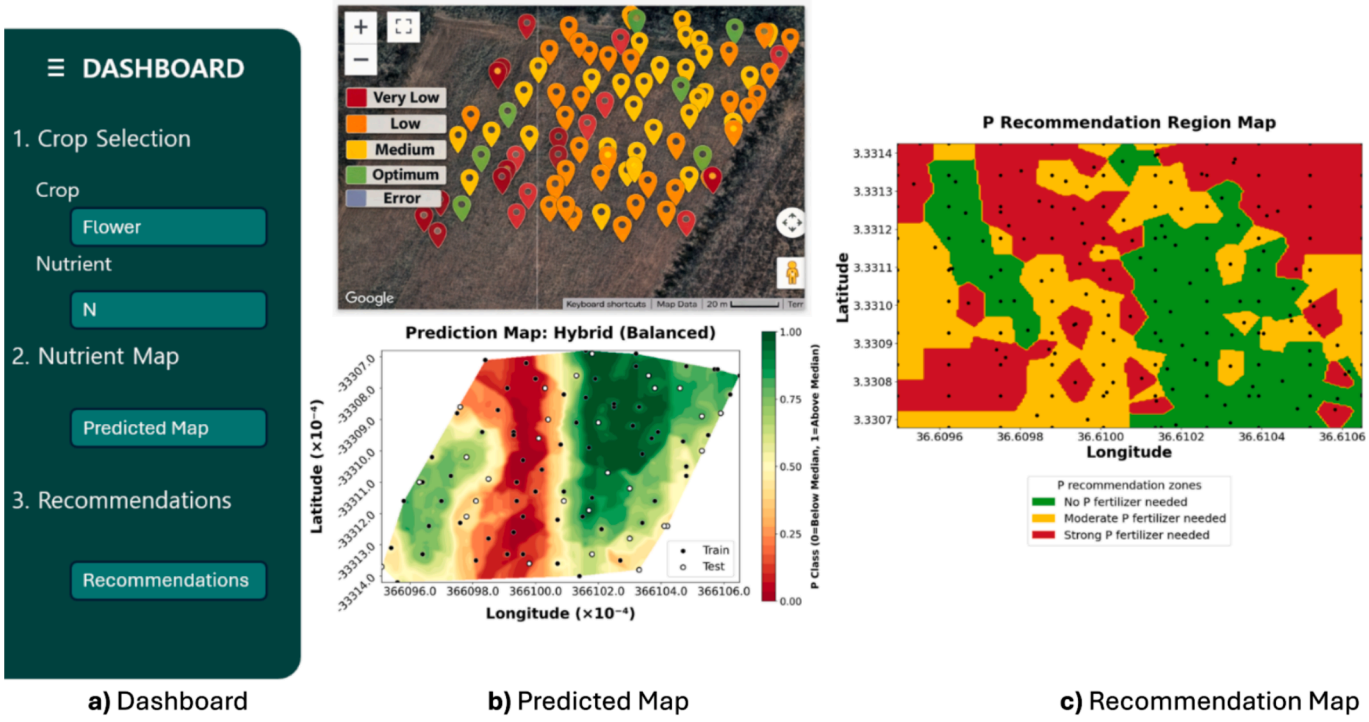


Fig. 11. GUI example with fertilizer recommendation map. This figure shows an example layout of the graphical user interface, including three main components: (a) the dashboard menu, (b) the predicted nutrient map, and (c) the fertilizer recommendation map. The left panel (a) displays the dashboard menu with options for crop selection, nutrient selection, nutrient map viewing, and fertilizer recommendations. The top center panel (b) shows field level nutrient indicators over a satellite image, with colored markers representing different nutrient categories. The bottom center panel (b) presents a soft classification map that displays phosphorus levels on a continuous scale. The right panel (c) shows a three-zone fertilizer recommendation map using tertile based classes, with each color representing a recommendation category.

used two classes and only one reading per location after aggregation. Study 3 used the full dataset and split phosphorus into three classes. Even though this increased the difficulty, accuracy and macro F1-score in Table 3b remained similar or slightly higher than in Study 2. The main change is that uncertainty shifted toward the medium class. Fig. 9c shows very strong performance for the low and high classes, while the medium class is less consistent, with an F1-score of 0.76. The confusion matrix in Fig. 10c illustrates this clearly. Errors mostly occur between low and medium or medium and high, rather than between the extremes. This makes sense because medium phosphorus values often sit near class boundaries. They also show more overlap with their neighbouring classes than the clear low and high regions. In other words, adding a third class helps capture more detail in the field but also introduces regions where the model has less certainty.

The feature importance patterns in Table 6 support this interpretation. When multiple readings per location were available, the model began relying more on location-level information. This is why location_mean_P ranked highest in Study 3. P_diff_from_neighbors and P_deviation_from_location_mean continued to matter, but they now worked alongside features such as is_location_min and is_location_max. These additional indicators helped the model decide which class a reading belonged to when values were close together. Spatial terms like longitude and lon_squared still appeared, but they played a slightly smaller role than in Study 2. This shift shows that the model takes advantage of repeated measurements by first anchoring readings to their local average, then refining the decision using neighbourhood and

spatial cues.

The broader comparison in Supplementary Table 3 shows that all three studies achieved similar mean cross-validation accuracies, with Study 3 being the most stable based on its narrow fold accuracy range. This tightening of variability from Study 1 to Study 3 suggests that clearer nutrient patterns and richer local information both help the model generalize more consistently. The confusion matrices and class-by-class scores across Fig. 7c, Fig. 8c and Fig. 9c follow a similar trend. Study 1 shows more symmetrical errors; Study 2 reduces them thanks to the stronger gradient and Study 3 concentrates them in the intermediate class.

From an agronomic point of view, the results show that the modelling approach is reliable at identifying areas that are clearly low or clearly high in phosphorus. These zones appear consistently across all three studies and show strong classification performance. The more ambiguous areas sit between these extremes, particularly in the three-class Tanzania case. These regions are still useful from a decision-making perspective because they highlight where additional sampling or expert judgment is needed. The stronger and more consistent patterns seen in Fig. 11 suggest that, across both fields and all label designs, the models produce maps and recommendation zones that can support fertilizer planning at a practical level while still leaving room for local agronomic insight.

The choice to formulate the problem as a classification task rather than a continuous regression also plays an important role in interpreting these results. The electrochemical probe used in this study provides indicative measurements that reflect relative nutrient availability rather than fully calibrated absolute concentrations. These readings are influenced by soil moisture, pH, ionic interference, and local soil matrix conditions, which introduce variability that is difficult to represent reliably as a continuous function without extensive site-specific calibration.

From a practical standpoint, agriculture decisions are typically based on identifying nutrient management zones rather than exact concentration values. The classification framework therefore aligns more directly with how fertilizer recommendations are applied in the field. In addition, classification reduces sensitivity to measurement noise by focusing on relative differences and spatial structure, which contributes to the stability observed across all three studies. While regression-based approaches could be explored in future work with larger calibrated datasets, the present results show that classification provides a robust and agronomically meaningful representation of spatial phosphorus variability under real field conditions.

7. Conclusion and future directions

This study demonstrated that UAV-enabled electrochemical soil sensing, when tightly integrated with spatially informed machine learning, can deliver field-scale phosphorus information with spatial detail difficult to achieve through conventional laboratory testing or imaging-based inference alone. The primary contribution is not a standalone algorithm, but a complete sensing-to-inference framework that couples autonomous soil probing, in situ nutrient measurement, GPS-referenced sampling, and spatial Random Forest classification into a single operational pipeline. Across three study designs and two contrasting field sites, this integrated system generated nutrient maps that were coherent, interpretable, and aligned with known agronomic patterns. The approach consistently identified low-phosphorus zones in the British Columbia vineyard and high-phosphorus zones in the Tanzanian flower farm, while also recovering broader spatial gradients even when the underlying nutrient range was narrow. Results from the full Tanzania dataset further showed that incorporating within-location readings allowed the system to represent micro-scale variability without sacrificing cross-validation stability.

System development and field deployment highlighted several practical constraints that are inseparable from real-world nutrient sensing. High mechanical loads during probe insertion in compact soils occasionally affected UAV stability, and early linkage designs introduced lateral forces that reduced insertion accuracy. Ground detection using the IR sensor was inconsistent under strong sunlight, and reliable electrochemical measurements required adequate soil moisture at the point of insertion. Field operations in grassy environments exposed takeoff and landing challenges from propeller obstruction. These observations motivate clear hardware refinements, including improved landing gear, soil pre-wetting mechanisms, and more robust ground-sensing strategies, all of which directly influence the quality and reliability of downstream model predictions. In addition, operational throughput remains constrained at the current prototype stage by the sensing cycle duration and payload-limited flight time. Scaling to larger agricultural areas will require either improved single-platform efficiency or distributed deployment strategies using multiple UAVs or larger carrier platforms. Addressing these factors is essential for scaling the approach beyond controlled trials and into routine agricultural use.

Modeling-wise, uncertainty concentrated in intermediate phosphorus levels, especially the three-class formulation. This reflects both the transitional nature of medium nutrient states and the sensitivity of electrochemical measurements to soil moisture, density, and insertion depth. Limited availability of standardized, crop-specific agronomic references constrained translation of sensor outputs into precise prescriptions. These challenges thus reinforce treating sensing and modeling as a coupled system rather than independent components. Future work should prioritize larger, more diverse field datasets, integration of external soil-health databases, and signal-filtering strategies that explicitly account for sensing noise. Hybrid approaches combining data-driven predictions with established agronomic rules offer a practical path toward safer and more actionable fertilizer guidance.

More broadly, results show spatial context as a primary signal for nutrient mapping, not a secondary refinement post-sensing. Features capturing local contrast, positional geometry, and within-location structure proved central to performance across sites and designs. This underscores that low-cost electrochemical sensors can be effective in environments where traditional interpolation struggles, provided that the physical sampling strategy and spatial modeling are designed together. The demonstrated sensing-to-modeling pipeline therefore provides a foundation for integrated decision-support systems, combining nutrient maps with yield data, imagery, weather records, and management history. In this sense, the work represents a step toward operational digital field representations that evolve with new measurements.

This work introduces a unified sensing-to-inference framework that advances beyond prior approaches by integrating four key components into a single operational system. First, it demonstrates autonomous UAV-based soil probing, enabling direct physical interaction with the soil rather than passive observation. Second, it incorporates in-situ electrochemical sensing to obtain real-time nutrient measurements at the point of contact, addressing the limitations of indirect spectral estimation. Third, it applies spatially informed machine learning to transform sparse, georeferenced measurements into continuous nutrient maps that capture both local variability and broader field-scale structure. Finally, these elements are combined into an end-to-end pipeline, linking hardware deployment, data acquisition, wireless communication, and spatial inference within a single workflow. This integration distinguishes the present study from existing work that typically addresses sensing, deployment, or modeling in isolation, and establishes a practical foundation for scalable, real-time nutrient monitoring in precision agriculture.

Overall, this work establishes a scalable approach bridging direct physical sensing and spatial inference via deliberate hardware–algorithm integration. By validating both the mechanical feasibility of autonomous soil probing and the analytical value of spatially informed learning, the study moves beyond algorithm-centric solutions and toward deployable systems. Continued refinement of sensing hardware

and modeling components will enable the framework to support precise fertilizer management, reduce input waste, and contribute sustainable agricultural practices across diverse environments.

Author contributions

W.S.K. conceived the methodological framework for AI-driven drone–soil sensing and supervised the project. D. L., A. A., M. A., T. J. S., D. B., J. M., and A. H. designed the mechatronic system on the drone. H.M. and P.B. conducted the experiments and interpreted the AI data analysis results.

CRediT authorship contribution statement

P. Biswas: Writing – original draft, Methodology. **H. Mohammadi:** Writing – original draft, Methodology. **D. Law:** Visualization, Methodology. **A. Al-Baidhani:** Methodology. **M. Abdirahman:** Methodology. **T.J. Suarez:** Methodology. **D. Bissada:** Methodology. **J. Musico:** Methodology. **A. Haq:** Methodology. **W.S. Kim:** Writing – review & editing, Methodology, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Woo Soo Kim reports financial support was provided by Simon Fraser University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper].

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compag.2026.111745.

Data availability

Data supporting the findings of this study are available within the manuscript. All other relevant data are available from authors upon reasonable request.

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